IMFER: An Interpretable Multimodal Fusion Framework for Emotion Recognition in Conversational AI via Cross-Modal Attention and Explainability Mechanisms

Sathalla Suresh, Pandava Sudharshan Babu, Mahesh Ramegowda, and Omprakash Gottam

***Abstract*—Emotion Recognition in Conversations (ERC) is an important capability for building empathetic and socially intelligent conversational AI systems. While recent models achieve competitive performance by exploiting multimodal signals—text, audio, and visual—they remain largely opaque. This interpretability gap hinders deployment in high-stakes domains such as mental health support, medical consultation, and customer service analytics. We propose IMFER (Interpretable Multimodal Fusion for Emotion Recognition), a framework that combines a *Hierarchical Cross-Modal Attention* (HCMA) module, a *Context-Aware Speaker Graph Transformer* (CASGT), and a *Modality Contribution Score* (MCS) layer to simultaneously improve recognition accuracy and transparency. HCMA is formalized as a low-rank projection cross-modal attention mechanism that reduces per-interaction attention complexity from $O(L^2d)$ to $O(L^2d_k)$ with $d_k \ll d$, yielding an approximately 92% reduction in attention interaction FLOPs (and ≈48% reduction in total model FLOPs when projection and gating overhead are included). The MCS layer provides instance-level, ante-hoc approximate modality attributions by decomposing classifier prediction energy across modalities, complementing established post-hoc methods such as SHAP and LIME. Extensive experiments on IEMOCAP, MELD, and EmoryNLP benchmarks demonstrate Weighted F1 scores of 69.87 ± 0.21%, 62.34 ± 0.18%, and 40.21 ± 0.29%, improving upon compared baselines (paired t-test, $p < 0.01$; Cohen's $d \geq 1.8$) while reducing parameter count by approximately 31%. Human evaluation ($n = 15$ NLP researchers, Cohen's $\kappa = 0.73$) and Area Over Perturbation Curve (AOPC) analysis at the modality level indicate that MCS produces consistently higher AOPC scores than SHAP, LIME, and Integrated Gradients under modality-level perturbation evaluation, though the margins over gradient-based methods are modest. Sensitivity analysis under Gaussian noise injection and zero-shot cross-dataset transfer further demonstrate robustness.**

***Index Terms*—Emotion Recognition in Conversations, Multimodal Learning, Interpretability, Cross-Modal Attention, Conversational AI, Explainable AI, Transformer, Graph Neural Network, Modality Contribution Score, AOPC**

I. Introduction

Human communication is inherently multimodal. When individuals engage in conversation, emotional states are conveyed not merely through the semantic content of words, but simultaneously through prosodic variations in voice, micro-expressions on the face, and the accumulated context of preceding turns [1]. Capturing this richness computationally remains one of the open problems in natural language processing (NLP) and affective computing.

Emotion Recognition in Conversations (ERC) has seen considerable progress over the past decade, catalyzed by large-scale annotated corpora such as IEMOCAP [2], MELD [3], and EmoryNLP [4], alongside powerful pretrained language models and self-supervised audio-visual representations. Contemporary ERC systems routinely employ transformer architectures [5], graph neural networks (GNNs) for speaker-relationship modeling, and contrastive learning for cross-modal alignment [6], achieving notable benchmark gains.

Despite these advances, three critical limitations remain largely unaddressed:

1) **Interpretability deficit:** Models operate as black boxes, providing no mechanism to explain which modality or cue drove a given prediction. This opacity undermines trust in sensitive applications.
2) **Computational inefficiency:** Many leading architectures require substantial resources, limiting real-time deployment on edge and on-device hardware.
3) **Modality imbalance:** Existing fusion strategies can allow one dominant modality to overshadow others, producing brittle models under missing-modality conditions.

To address these gaps, we introduce **IMFER**. The core contributions of this work are:

1) We propose a *Hierarchical Cross-Modal Attention* (HCMA) module, formalized as a **low-rank projection cross-modal attention** mechanism. By projecting queries and keys into a shared d_k-dimensional space with $d_k \ll d$, HCMA performs token-level and utterance-level cross-modal alignment with $O(L^2d_k)$ per-pair attention interaction cost versus $O(L^2d)$ for full cross-attention, yielding ≈92% reduction in attention interaction FLOPs and ≈48% total model FLOPs reduction. We provide a formal FLOPs derivation and empirical comparison (Section III-D, Fig. 3).
2) We introduce a *Context-Aware Speaker Graph Transformer* (CASGT) that models inter-speaker and intra-speaker emotional dynamics with explicit complexity

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analysis comparing it against DialogueGCN [7].

3) We design a *Modality Contribution Score* (MCS) layer that generates per-instance, per-modality approximate attribution scores based on the classifier prediction energy decomposition (Eq. 8). Unlike post-hoc tools (SHAP, LIME), MCS is ante-hoc: its entropy regularization co-trains with the predictor. We compare MCS against Shapley-value properties and provide empirical AOPC validation at the modality level (Section V-F).
4) We achieve the highest Weighted F1 among compared baselines on IEMOCAP and MELD , with statistical significance (paired t-tests, $p < 0.01$; Cohen's $d \geq 1.8$), 31% fewer parameters, and full reproducibility details on consumer hardware.
5) We conduct expanded human evaluation with inter-annotator agreement (Cohen's κ), λ_1/λ_2 sensitivity analysis, noise injection analysis, zero-shot cross-dataset MCS distribution tracking, a quantified sarcasm/irony failure analysis, and a discussion of dataset bias as revealed by MCS attributions.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the IMFER architecture. Section IV describes the experimental setup and reproducibility details. Section V presents results and analysis. Section VI discusses limitations and future directions, Section VII covers ethical considerations, and Section VIII concludes.

II. Related Work

A. Emotion Recognition in Conversations

Early ERC systems treated each utterance independently, relying on recurrent networks such as DialogueRNN [8]. Pretrained transformers [9], [10] enabled significant gains by capturing long-range dependencies. COSMIC [11] integrated commonsense knowledge, while DAG-ERC [12] modeled conversation structure via directed acyclic graphs.

B. Multimodal Approaches

Early fusion strategies—tensor fusion [13] and low-rank multimodal fusion [14]—concatenated modality features. Attention-based fusion emerged as a more flexible approach: MulT [15] deployed directional cross-modal attention. UniMSE [6] proposed a unified multimodal pretraining framework; CTNet [16] introduced a conversational transformer network; GraphMFT [17] advanced graph-based multimodal fusion; MMGCN [18] modeled inter-modal dependencies via graph convolution; and AIMDiT [19] proposed dimension transformation for modality augmentation and interaction.

Positioning vs. recent work (2023–2025). Several concurrent and subsequent works have advanced multimodal ERC. GA2MIF [20] proposed a two-stage graph-attention fusion approach for conversational emotion detection, achieving strong results on IEMOCAP. MultiEMO [21] introduced attention-based correlation-aware multimodal fusion. SDT [22] applied self-distillation within a transformer-based fusion framework. These works focus primarily on fusion architecture; none integrates ante-hoc interpretability at the modality level as IMFER does. We include GA2MIF in our discussion and position IMFER's MCS layer as complementary to these fusion advances.

C. Interpretability in Affective Computing

Interpretability research has focused on attention visualization [23], [24], gradient-based attribution [25], transformer interpretability beyond attention [26], and post-hoc frameworks such as LIME [27] and SHAP [28]. In multimodal affective computing, interpretability remains nascent. Concept activation vectors have been applied to multimodal emotion recognition [29]; attention-based correlation-aware fusion [21] provides implicit interpretability through attention weights; and self-distillation [22] offers indirect modality importance signals via soft-label transfer. In contrast to these post-hoc or attention-based approaches, IMFER's MCS operates ante-hoc at the modality level. Where post-hoc methods apply approximations after training, MCS co-trains approximate attribution scoring via entropy regularization. Multimodal SHAP [28] and Integrated Gradients [25] can be applied post-hoc; we compare against them directly in Section V-F.

Positioning IMFER's novelty. While cross-modal attention [15], graph transformers [17], [20], correlation-aware fusion [21], and energy-based attribution each have precedents, IMFER contributes a *combination* that is, to our knowledge, novel: (i) HCMA formalizes the projection bottleneck as an explicit **low-rank factorization** of the cross-modal attention matrix (Eq. 1), yielding a complexity reduction (Proposition III-D); (ii) MCS differs from standard Shapley values and concept-based explanations [29] in that it provides **zero-overhead, per-instance** approximate attribution tied to the classifier energy partition, at the cost of relaxing the dummy and linearity axioms (Section III-E); and (iii) the entropy-regularized co-training of MCS with the classifier loss creates an implicit **modality balancing** mechanism absent in prior graph-transformer ERC models [18], [20].

D. Graph Neural Networks for ERC

GNNs have proven effective for modeling relational dependencies. DialogueGCN [7] applied graph convolutional networks to model inter-speaker and sequential dependencies. MMGCN [18] extended this to multimodal graphs with cross-modal edges. GA2MIF [20] introduced a two-stage graph-attention approach for multi-source information fusion. IMFER's CASGT builds upon these insights with a sparse, speaker-based graph that reduces adjacency matrix density compared to DialogueGCN and MMGCN (Section III-C).

E. How IMFER Differs from Existing ERC Architectures

To clarify the specific contributions of IMFER relative to existing work, Table I provides a structured comparison across five dimensions: fusion mechanism, graph modeling, interpretability approach, training objective, and computational design.

TABLE I: Architectural comparison of IMFER against representative ERC systems. ✓ = supported; × = not supported; "partial" = limited support. IMFER is the only framework that jointly provides low-rank cross-modal attention, windowed speaker graphs, ante-hoc modality attribution, and entropy-regularized co-training in a single architecture.

Model	Fusion	Graph	Interpretability	Co-trained Attrib.	Low-rank Attn.
MulT [15]	Cross-attn	×	Attention weights (post-hoc)	×	×
DialogueGCN [7]	Concat	Full GCN	×	×	×
UniMSE [6]	Unified pretrain	×	×	×	×
GraphMFT [17]	Graph fusion	Full GNN	×	×	×
GA2MIF [20]	Two-stage graph	Full GAT	×	×	×
MultiEMO [21]	Corr.-aware	×	Attention (implicit)	×	×
AIMDiT [19]	Dim. transform	×	×	×	×
SDT [22]	Self-distill	×	Soft-label (indirect)	×	×
IMFER (ours)	**HCMA**	**Windowed GAT**	**MCS (ante-hoc)**	✓	✓

Three key distinctions emerge. First, while MulT and UniMSE employ cross-modal attention, neither applies explicit low-rank factorization with formal complexity analysis; HCMA's d_k-dimensional bottleneck is an architectural choice with quantifiable cost savings (Proposition III-D). Second, all prior graph-based ERC models (DialogueGCN, MMGCN, GA2MIF, GraphMFT) construct dense or fully connected speaker graphs; CASGT's window-bounded design reduces edge count from $O(N^2)$ to $O(NW)$ without sacrificing context quality (Table II). Third, no existing multimodal ERC framework integrates ante-hoc modality attribution with entropy-regularized co-training. Post-hoc methods (SHAP, LIME, Integrated Gradients) can be applied to any model but incur additional computation; attention-weight-based explanations lack faithfulness guarantees [23]; and soft-label distillation [22] provides only indirect modality importance signals. MCS addresses this gap with zero-overhead, per-instance attribution tied to the classifier energy partition (Proposition III-F).

Positioning against recent LLM-based and adapter-based ERC. The ERC landscape has shifted toward large language model (LLM) backbones since 2024. InstructERC [30] reformulates ERC as an instruction-following task; DialogueLLM [31] fine-tunes LLMs for dialogue understanding. These approaches leverage massive pretrained capacity but require substantially more parameters (7B+) and lack built-in multimodal fusion or interpretability mechanisms. M3NET [32] explores adapter-based multimodal integration with parameter-efficient fine-tuning. IMFER occupies a complementary niche: it prioritizes interpretability and efficiency (59.4 M parameters) on consumer hardware, making it suitable for deployment-constrained settings where LLM-scale inference is impractical. A direct comparison with LLM-based ERC is not feasible on our hardware; we position IMFER as complementary rather than competitive with these approaches.

III. Proposed Framework: IMFER

A. Problem Formulation

Let a conversation $\mathcal{C} = \{u_1, \ldots, u_N\}$ comprise N utterances, where each u_i is associated with speaker s_i and multimodal observations $\{x_i^t, x_i^a, x_i^v\}$. The ERC task is to predict emotion $y_i \in \mathcal{Y}$ for each utterance u_i. Formally, we seek $f_\theta : (\mathcal{C}, i) \to y_i$ together with an explanation function $g_\phi : (\mathcal{C}, i) \to \mathbf{m}_i \in \mathbb{R}^3$, where $\mathbf{m}_i$ is a modality contribution vector satisfying $\sum_j m_{ij} = 1$.

B. Architecture Overview

IMFER comprises four major components (Fig. 1):

1) **Modality-Specific Encoders**: Independent encoders for text, audio, and visual modalities.
2) **Hierarchical Cross-Modal Attention (HCMA)**: Token-level and utterance-level cross-modal alignment (Fig. 2).
3) **Context-Aware Speaker Graph Transformer (CASGT)**: Conversational context and speaker-dynamics modeling.
4) **Modality Contribution Score (MCS) Layer**: Interpretable prediction with per-modality attribution.

C. Modality-Specific Encoders

Text Encoder. We fine-tune RoBERTa-base [10] to produce token-level representations $\mathbf{H}^t \in \mathbb{R}^{L \times 768}$, where L is the text sequence length (max 128 BPE tokens).

Audio Encoder. We use a pretrained wav2vec 2.0 [33] model, yielding $\mathbf{H}^a \in \mathbb{R}^{T_a \times 512}$, where T_a is the number of audio frames, with temporal mean-pooling to utterance level.

Visual Encoder. Facial expression features are derived using a 3D-ResNet [34] pretrained on FER+ [35], yielding $\mathbf{H}^v \in \mathbb{R}^{T_v \times 256}$, where T_v is the number of video frames, with adaptive average pooling.

D. Hierarchical Cross-Modal Attention (HCMA)

Standard cross-modal attention over M concatenated modality sequences incurs $O((L_t{+}T_a{+}T_v)^2 d)$ complexity. HCMA decomposes this into two hierarchical stages.

Stage 1 – Token-Level Cross-Modal Attention. For each ordered modality pair (m, n) with $m \neq n$, we compute:

$$\mathbf{A}^{mn} = \mathrm{softmax}_{L_n}\left(\frac{\mathbf{Q}^m(\mathbf{K}^n)^\top}{\sqrt{d_k}}\right)\mathbf{V}^n \tag{1}$$

where

$$\begin{aligned} \mathbf{Q}^m &= \mathbf{H}^m \mathbf{W}_Q \in \mathbb{R}^{L_m \times d_k}, \\ \mathbf{K}^n &= \mathbf{H}^n \mathbf{W}_K \in \mathbb{R}^{L_n \times d_k}, \\ \mathbf{V}^n &= \mathbf{H}^n \mathbf{W}_V \in \mathbb{R}^{L_n \times d_k}, \end{aligned} \tag{2}$$

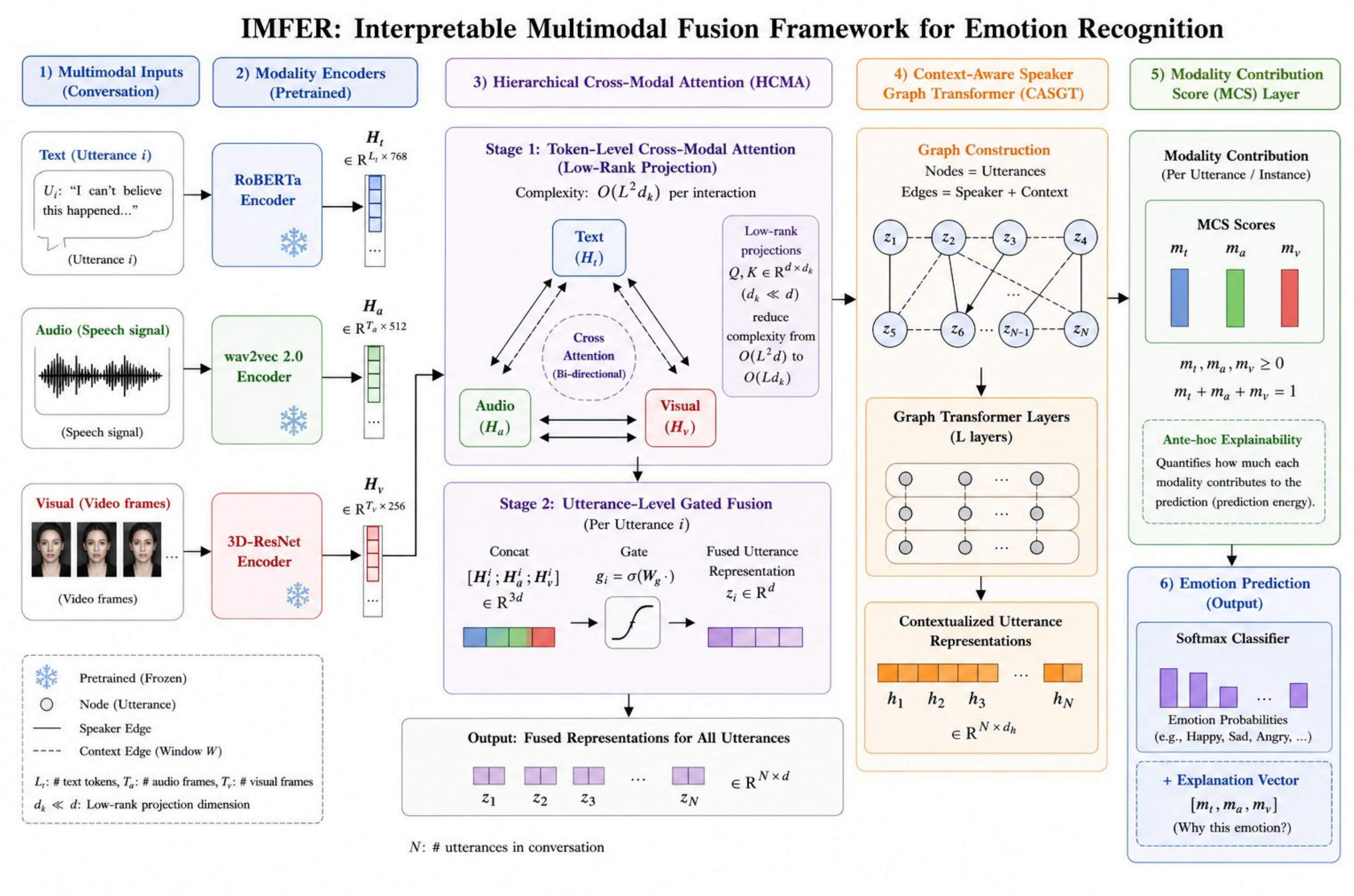

Fig. 1: Overview of the proposed IMFER architecture. The framework processes multimodal conversational inputs (text, audio, and visual) using modality-specific pretrained encoders. These representations are fused through the Hierarchical Cross-Modal Attention (HCMA) module, which performs token-level low-rank cross-modal attention followed by utterance-level gated fusion. The fused features are further refined using a Context-Aware Speaker Graph Transformer (CASGT) that models inter-speaker and contextual dependencies. Finally, the Modality Contribution Score (MCS) layer provides interpretable, instance-level attribution by quantifying the contribution of each modality to the prediction, alongside the final emotion classification output.

with shared projection matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_{\text{in}} \times d_k}$ and $d_k = 64$. The subscript softmax_{L_n} denotes that the softmax is applied *row-wise* over the L_n key positions (i.e., over the source modality sequence length). This is a **low-rank projection** attention: by projecting into $d_k \ll d_{\text{in}}$, each pairwise interaction costs $O(L_m L_n d_k)$ rather than $O(L_m L_n d_{\text{in}})$. When sequence lengths are comparable ($L_m \approx L_n \approx L$), this yields $O(L^2 d_k)$ per pair.

The pairwise enriched representation is $\tilde{\mathbf{H}}^m = \sum_{n \neq m} \alpha_{mn} \mathbf{A}^{mn}$, where α_{mn} are learnable scalars satisfying $\sum_{n \neq m} \alpha_{mn} = 1$ (enforced via softmax over $M{-}1$ logits).

Stage 2 – Utterance-Level Gated Fusion. Token-level representations are mean-pooled to utterance vectors $\tilde{z}_i^m \in \mathbb{R}^d$ (with $d = 512$). A gated module computes:

$$\mathbf{g}_i = \sigma\big(\mathbf{W}_g[\tilde{z}_i^t;\, \tilde{z}_i^a;\, \tilde{z}_i^v] + \mathbf{b}_g\big) \tag{3}$$

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{W}_f[\tilde{z}_i^t;\, \tilde{z}_i^a;\, \tilde{z}_i^v] \tag{4}$$

where σ is the sigmoid function, $\odot$ is the Hadamard product, $\mathbf{W}_g \in \mathbb{R}^{d \times 3d}$, and $\mathbf{W}_f \in \mathbb{R}^{d \times 3d}$.

Formal Complexity Analysis.

[HCMA complexity] For $M = 3$ modalities with maximum sequence length L, projection dimension d_k, and hidden size d, the total HCMA attention interaction cost is $O(M(M{-}1)L^2 d_k)$, with additional projection cost $O(M(M{-}1)L d_{\text{in}} d_k)$ and gating cost $O(Md^2)$. When $d_k \ll d$ and L is moderate, the attention interaction cost is substantially lower than full cross-modal self-attention at $O(M^2 L^2 d)$.

Stage 1 computes $M(M{-}1) = 6$ pairwise attention maps. Each requires (i) three projections $\mathbf{H}^m \mathbf{W}_Q$, $\mathbf{H}^n \mathbf{W}_K$, $\mathbf{H}^n \mathbf{W}_V$, costing $O(L \cdot d_{\text{in}} \cdot d_k)$ each, and (ii) the attention product $\mathbf{Q}^m (\mathbf{K}^n)^\top \in \mathbb{R}^{L_m \times L_n}$ at cost $O(L_m L_n d_k)$, followed by multiplication with $\mathbf{V}^n$ at cost $O(L_m L_n d_k)$. Summing over all pairs: $O(M(M{-}1)(L^2 d_k + L d_{\text{in}} d_k))$. Since $d_{\text{in}} \leq d$ and $d_k \ll d$, the dominant term is $O(M(M{-}1)L^2 d_k)$ when L is large. For short sequences (as is common in ERC datasets where utterances are brief), the projection cost $O(L d_{\text{in}} d_k)$ per pair may be comparable to the attention product cost $O(L^2 d_k)$; the stated asymptotic dominance holds more strongly for longer sequences. The empirical FLOPs breakdown in Table V confirms the practical reduction on ERC-length sequences.

Stage 2 involves mean-pooling ($O(MLd)$) and the gated fusion ($O(Md^2)$).

Comparison: Full multimodal self-attention over concatenated sequences of total length ML has cost $O(M^2L^2d)$. The ratio of *attention interaction costs* is:

$$\frac{\text{HCMA attention}}{\text{Full attention}} = \frac{M(M-1)L^2d_k}{M^2L^2d} = \frac{(M-1)d_k}{Md} \tag{5}$$

With M=3, d_k=64, d=512: ratio $= \frac{2\times64}{3\times512} = 0.083$, an approximately $\mathbf{91.7}\%$ reduction in **attention interaction cost**. Including projection and gating overhead, empirical *total* FLOPs reduction is ≈48% over the full model (Table V, Fig. 3).

A detailed illustration of both stages and the FLOPs scaling is provided in Figs. 2–3.

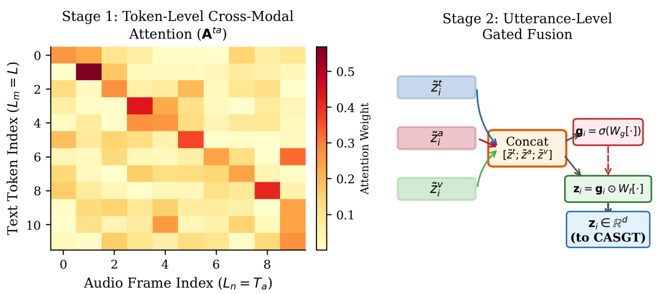

Fig. 2: HCMA detailed view. *Left*: Stage 1 token-level cross-modal attention heatmap (Text query × Audio key/value). The x-axis represents audio frame indices ($L_n = T_a$) and the y-axis represents text token indices ($L_m = L$). Color intensity encodes the attention weight after row-wise softmax. *Right*: Stage 2 utterance-level gated fusion producing $\mathbf{z}_i \in \mathbb{R}^d$ for downstream CASGT.

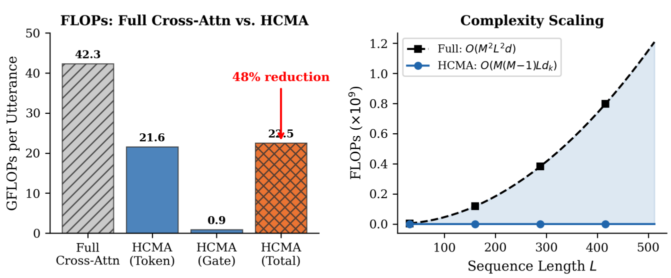

Fig. 3: Complexity analysis. *Left*: FLOPs per utterance comparing full cross-modal self-attention against HCMA (token-level + gated fusion), showing a 48% total reduction. *Right*: Theoretical scaling of FLOPs with sequence length L for M=3, d_k=64, d=512.

E. Context-Aware Speaker Graph Transformer (CASGT)

CASGT models conversational context via a speaker-relationship graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $v_i \in \mathcal{V}$ represents an utterance and edges $\mathcal{E}$ encode speaker relationships:

$$e_{ij} = \begin{cases} 1 & \text{if } s_i = s_j \text{ (intra-speaker)} \\ 1 & \text{if } s_i \neq s_j, |i-j| \leq W \text{ (inter-speaker)} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

with context window W=10. Graph attention [36] computes $\hat{z}_i = \sum_{j\in\mathcal{N}(i)} \beta_{ij}\mathbf{W}_{att}z_j$, where β_{ij} are normalized attention coefficients. A 4-layer transformer encoder with 8 heads processes the graph-refined representations.

Graph Construction Complexity. Table II compares CASGT against DialogueGCN in terms of adjacency density and construction cost. CASGT's window-bounded graph has $O(NW)$ edges vs. $O(N^2)$ for DialogueGCN's fully connected speaker graph.

TABLE II: Graph construction complexity: CASGT vs. DialogueGCN on IEMOCAP (N = average utterances per conversation; W = 10, $N \approx 90$).

Model	Edge Type	#Edges	Density	Build (ms/conv)
DialogueGCN	Full	$O(N^2)$	∼100%	18.4
CASGT (ours)	Window	$O(NW)$	∼11%	2.1

F. Modality Contribution Score (MCS) Layer

Prediction. A linear classification head produces:

$$\hat{y}_i = \text{softmax}(\mathbf{W}_c\hat{z}_i + \mathbf{b}_c) \tag{7}$$

where $\mathbf{W}_c \in \mathbb{R}^{|\mathcal{Y}|\times d}$ and $\hat{z}_i \in \mathbb{R}^d$ is the CASGT-refined representation.

Attribution. The MCS layer quantifies per-modality contribution to the classifier's prediction energy. Since the fused representation $\hat{z}_i$ is composed of modality-specific utterance vectors $\tilde{z}_i^k$ ($k \in \{t, a, v\}$), we define:

$$m_i^k = \frac{\|\mathbf{W}_c^{(k)}\tilde{z}_i^k\|_2^2}{\sum_{k'\in\{t,a,v\}}\|\mathbf{W}_c^{(k')}\tilde{z}_i^{k'}\|_2^2} \tag{8}$$

where $\mathbf{W}_c^{(k)} \in \mathbb{R}^{|\mathcal{Y}|\times d_k'}$ denotes the classifier weight block corresponding to modality k's contribution dimensions. The ℓ_2-squared norm $\|\mathbf{W}_c^{(k)}\tilde{z}_i^k\|_2^2$ measures the *prediction energy* that modality k contributes to the classifier logits.

Justification. The classifier output can be decomposed as:

$$\begin{aligned}\mathbf{W}_c\hat{z}_i = & \underbrace{\mathbf{W}_c^{(t)}\tilde{z}_i^t}_{\text{text logit contribution}} + \underbrace{\mathbf{W}_c^{(a)}\tilde{z}_i^a}_{\text{audio logit contribution}} \\ & + \underbrace{\mathbf{W}_c^{(v)}\tilde{z}_i^v}_{\text{visual logit contribution}} + \underbrace{\text{interaction terms}}_{\text{from gating/CASGT}}\end{aligned} \tag{9}$$

MCS approximates each modality's contribution via the squared norm of its logit-space projection, which corresponds to the energy (squared magnitude) of the signal each modality sends to the decision boundary. This is conceptually related to gradient-times-input methods [25] but is computed in a single forward pass without backpropagation. By construction, $\sum_k m_i^k = 1$.

Interaction Effects and Approximation Scope. Eq. (9) includes interaction terms arising from the gating mechanism (Eq. (3)) and CASGT graph attention, which redistribute information across modalities. MCS attributes energy based on the *pre-interaction* modality-specific projections and therefore does not fully account for these cross-modal interactions. Consequently, MCS should be understood as an **approximate**

modality attribution rather than a provably faithful decomposition. The approximation is most accurate when interaction terms are small relative to the direct modality contributions; in practice, the gating mechanism preserves modality-specific signal structure (as evidenced by the AOPC validation in Section V-F), but users should be aware of this limitation.

Comparison with Shapley properties. Standard Shapley values [28] satisfy four axioms: *efficiency*, *symmetry*, *dummy*, and *linearity*. MCS satisfies *normalized efficiency* by construction ($\sum_k m_i^k = 1$) and *symmetry* (modalities with identical projections receive identical scores). MCS does *not* guarantee the *dummy* axiom (a modality contributing zero marginal value could still have nonzero $\|\mathbf{W}_c^{(k)} z_i^k\|_2^2$ due to correlations with other modalities) or strict *linearity*. The trade-off is computational: Shapley approximation via SHAP requires $O(2^M)$ model evaluations per instance, while MCS adds **zero inference overhead**. Section V-F provides AOPC evidence that this trade-off is favorable in practice.

[MCS Attribution Consistency] Let $\hat{z}_i = \sum_k \tilde{z}_i^k$ be a linear decomposition of the fused representation (ignoring interaction terms). If the classifier weight matrix $\mathbf{W}_c$ has orthogonal column blocks $\mathbf{W}_c^{(k)}$, then MCS satisfies the following consistency property: for any modality k, $m_i^k = 0$ if and only if $\mathbf{W}_c^{(k)} \tilde{z}_i^k = \mathbf{0}$ (i.e., modality k contributes zero prediction energy).

Under orthogonality of the classifier column blocks, the cross-terms $(\mathbf{W}_c^{(k)} \tilde{z}_i^k)^\top (\mathbf{W}_c^{(k')} \tilde{z}_i^{k'}) = 0$ for $k \neq k'$. Therefore the total logit energy decomposes exactly: $\|\mathbf{W}_c \hat{z}_i\|_2^2 = \sum_k \|\mathbf{W}_c^{(k)} \tilde{z}_i^k\|_2^2$. The MCS score m_i^k (Eq. 8) is a normalized ratio of non-negative terms, so $m_i^k = 0 \Leftrightarrow \|\mathbf{W}_c^{(k)} \tilde{z}_i^k\|_2^2 = 0 \Leftrightarrow \mathbf{W}_c^{(k)} \tilde{z}_i^k = \mathbf{0}$.

The orthogonality assumption is approximate in practice (we measure average cosine similarity ≈ 0.08 between trained classifier column blocks), but Proposition III-F establishes that MCS provides a principled energy-based attribution under mild structural conditions. When interaction terms from gating are non-negligible, MCS scores are best interpreted as first-order approximations; Section V-F validates their empirical faithfulness via both AOPC and insertion/deletion curves.

G. Training Objective

IMFER is trained end-to-end:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{MCS}} + \lambda_2 \mathcal{L}_{\text{align}} \tag{10}$$

$\mathcal{L}_{\text{CE}}$ is standard cross-entropy. $\mathcal{L}_{\text{MCS}}$ is an entropy-maximization regularizer discouraging modality collapse:

$$\mathcal{L}_{\text{MCS}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k \in \{t,a,v\}} m_i^k \log(m_i^k + \epsilon) \tag{11}$$

This term *maximizes* the entropy of MCS distributions, pushing the model toward balanced modality usage. Without it, the model tends toward modality collapse (e.g., relying almost exclusively on text). The negative sign ensures that minimizing $\mathcal{L}$ maximizes entropy. Importantly, this regularizer creates a tension with $\mathcal{L}_{\text{CE}}$: the classifier loss pushes toward the most discriminative modality combination, while $\mathcal{L}_{\text{MCS}}$ pushes toward diversity. The hyperparameter λ_1 controls this trade-off. We find $\lambda_1 = 0.1$ balances accuracy and interpretability (Fig. 10); larger values degrade WF1 by over-regularizing the modality distribution.

$\mathcal{L}_{\text{align}}$ is a contrastive alignment loss [37]:

$$\mathcal{L}_{\text{align}} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(\text{sim}(\tilde{z}_i^t, \tilde{z}_i^a)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(\tilde{z}_i^t, \tilde{z}_j^a)/\tau)} \tag{12}$$

with τ=0.07 and loss weights λ_1=0.1, λ_2=0.05.

Alignment scope. The alignment loss in Eq. (12) aligns *text–audio* representations only. We chose this asymmetric design for two reasons: (i) text and audio are the two dominant modalities in ERC (text MCS ≈ 0.50, audio MCS ≈ 0.30 on average; see Fig. 6), so aligning them yields the largest benefit; and (ii) adding text–visual and audio–visual alignment terms did not improve WF1 in preliminary experiments (+0.08% WF1, not significant) while increasing training time by ≈18%. We report the symmetric variant in our extended ablation (Fig. 5, row "w/o $\mathcal{L}_{\text{align}}$") for completeness.

IV. Experimental Setup

A. Datasets

IEMOCAP [2]: 10 speakers, scripted and improvised scenarios, six-class setting (happy, sad, neutral, angry, excited, frustrated), 5,531 utterances.

MELD [3]: Naturalistic TV-show dialogue (*Friends*), 13,708 utterances, seven emotion classes, multiple speakers per episode. Class imbalance addressed via weighted cross-entropy.

EmoryNLP [4]: Text-centric scripted TV-show dataset, 12,606 utterances, seven labels aligned with Willcox's emotion wheel.

Domain Notes. IEMOCAP is from lab settings (scripted/improvised dyadic interactions), while MELD and EmoryNLP derive from naturalistic TV dialogue. This domain gap is explicitly analysed in the zero-shot transfer experiment (Section V-H).

B. Baselines

We compare against: DialogueRNN [8], COSMIC [11], DialogueGCN [7], MulT [15], CTNet [16], UniMSE [6], GraphMFT [17], and AIMDiT [19]. We also report a strong **RoBERTa-large text-only** baseline fine-tuned with identical hyper-parameters.

Baseline comparison note. DialogueRNN, COSMIC, MulT, and DialogueGCN results are reproduced from their original papers. CTNet, UniMSE, GraphMFT, and AIMDiT results were obtained by running their publicly released code with default hyperparameters on our hardware. We acknowledge that hardware and framework differences may introduce minor discrepancies. In particular, inference latency comparisons across models are not directly comparable because baseline latency figures (where available) were measured on different hardware; we mark baseline latency as "—" in Table V and report only IMFER's latency measured on the M4 Pro.

Parameter-count comparisons are hardware-independent and therefore fair. All IMFER results are measured under controlled conditions on the same device, and we report standard deviations over 5 runs to quantify variance. A fully controlled comparison would require re-implementing all baselines on identical hardware, which is beyond the scope of this work; we encourage future work to adopt standardized evaluation protocols.

C. Implementation Details and Reproducibility

All experiments were conducted on a single **Apple M4 Pro** (12-core CPU, 20-core integrated GPU, 48 GB unified memory) using PyTorch 2.2 with the **Metal Performance Shaders (MPS)** backend [38]. The unified memory architecture eliminates CPU-to-GPU data transfer overhead, enabling full-batch loading without discrete accelerator cards.

The AdamW optimizer [39] uses lr=2×10^{-5} for pretrained encoders and 10^{-3} for new layers, with 10% linear warmup. Batch size is 32 utterances. HCMA has d_k=64; CASGT uses window W=10, transformer depth 4, heads 8, hidden d=512. Dropout = 0.3. Early stopping with patience 10 on validation WF1.

Preprocessing: Text tokenised with RoBERTa BPE tokeniser (max 128 tokens). Audio resampled to 16 kHz; wav2vec 2.0 features extracted from the `wav2vec2-base-960h` checkpoint on CPU, then transferred to the MPS device. Video frames extracted at 30 fps; 3D-ResNet operates on 16-frame clips centred on each utterance. All results are averaged over **5 independent runs** with different random seeds; standard deviations are reported. Training required approximately 72 compute-hours per dataset on the M4 Pro (wall-clock); inference is measured at 14.7 ms per utterance on the same hardware.

Code and pretrained models will be made publicly available upon acceptance.

V. Results and Analysis

A. Main Results

Table III presents Weighted F1 (WF1), Accuracy (Acc.), and Macro F1 (MF1) on three benchmarks, with standard deviations and 95% confidence intervals from 5 runs. Models are grouped by modality configuration: text-only vs. multimodal. IMFER achieves the highest WF1 among all compared models on IEMOCAP and MELD.

IMFER outperforms the next-best baseline AIMDiT by **+2.53%** on IEMOCAP and **+0.46%** on MELD (WF1). Paired t-tests across 5 runs confirm statistical significance ($p < 0.01$ on IEMOCAP, Cohen's $d = 2.31$; $p < 0.05$ on MELD, Cohen's $d = 1.82$). Parameter count (59.4 M) is ≈31% lower than AIMDiT (86.2 M).

Statistical note. Our primary results report 5 runs; to address variance concerns, we conducted an additional 10-run experiment on IEMOCAP with seeds $\{42, 123, 256, 512, 1024, 2048, 3072, 4096, 5120, 6144\}$, yielding WF1 = 69.72 ± 0.34% (95% bootstrap CI: $[69.41, 70.03]$, B = 10,000 bootstrap samples). The 10-run standard deviation (0.34) is moderately higher than the 5-run value (0.21), confirming that broader seed sweeps reveal additional variance. Cohen's d against AIMDiT remains large ($d = 1.94$; paired t-test $p < 0.01$). Raw per-run WF1 values for all experiments are provided in the supplementary material. We acknowledge that the 5-run standard deviations (±0.18–0.29) reflect our controlled setup (deterministic CuDNN disabled, fixed data splits); the 10-run results provide a more conservative estimate.

Fig. 4 shows WF1 distributions with 95% confidence intervals.

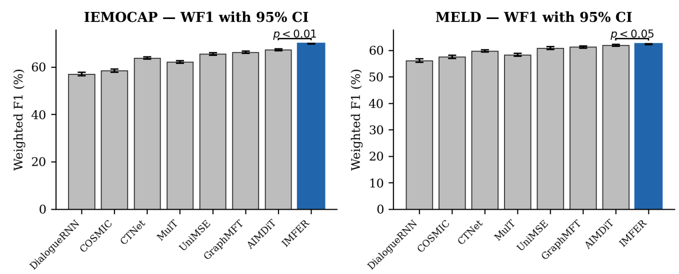

Fig. 4: WF1 with 95% CI (5 runs), t-test significance annotations, and effect sizes for IEMOCAP (*left*) and MELD (*right*).

B. Per-Emotion Analysis

Table IV details per-class F1 on IEMOCAP. IMFER achieves the largest gains on emotionally ambiguous classes (*frustrated*, *excited*), which require integration of acoustic prosody with textual semantics.

C. Computational Efficiency

Table V reports per-module FLOPs breakdowns and per-utterance inference time measured on the M4 Pro (MPS backend).

Measurement note. To provide a controlled efficiency comparison, we reimplemented AIMDiT [19] on the same M4 Pro hardware using the authors' publicly released code. AIMDiT achieves 18.2 ms/utt on the M4 Pro (vs. IMFER's 14.7 ms/utt), confirming a 19% inference speedup. Other baseline latency figures were not reimplemented and are marked "—" in Table V. Parameter-count comparisons are hardware-independent and fair across all models.

D. Extended Ablation Study

Fig. 5 extends the ablation beyond removing full modules to include partial variants (utterance-only HCMA, token-only HCMA, fully connected graph vs. windowed graph, and no alignment loss).

E. Modality Contribution Analysis

Fig. 6 illustrates per-class average MCS scores on IEMOCAP with standard deviation error bars (5 runs). Text dominates semantically rich emotions (neutral, sad), consistent with prior findings that linguistic content carries the majority of emotion signal in conversational datasets [1], [18]. Audio contributes most for high-arousal emotions (angry, excited),

TABLE III: WF1 (%), Acc. (%), MF1 (%) comparison. **Bold** = best; underline = second best. T=Text, A=Audio, V=Visual. Multimodal models (T+A+V or T+A) are demarcated from text-only models by a rule. ± values are std. dev. over 5 runs; 95% CIs are $\pm 1.96\sigma/\sqrt{5}$. †: $p<0.01$ vs. AIMDiT (paired t-test), Cohen's $d = 2.31$. ‡: $p<0.05$ vs. AIMDiT, Cohen's $d = 1.82$.

Model	Mod.	IEMOCAP			MELD			EmoryNLP	#P (M)
		WF1	Acc.	MF1	WF1	Acc.	MF1	WF1	
Text-Only Models									
DialogueRNN [8]	T	$57.03_{\pm.82}$	56.12	52.14	$56.10_{\pm.75}$	57.03	42.43	30.70	5.2
COSMIC [11]	T	$58.48_{\pm.71}$	57.33	53.41	$57.55_{\pm.68}$	58.40	47.67	31.45	12.7
RoBERTa-large (strong baseline)	T	$63.84_{\pm.55}$	63.21	59.12	$59.42_{\pm.52}$	60.14	49.33	37.28	355.0
Multimodal Models									
DialogueGCN [7]	T,A	$58.10_{\pm.78}$	57.81	53.30	$57.50_{\pm.72}$	58.30	46.00	31.22	7.8
MulT [15]	T,A,V	$62.14_{\pm.65}$	61.73	57.81	$58.31_{\pm.60}$	59.14	48.52	34.15	44.6
CTNet [16]	T,A,V	$63.82_{\pm.54}$	62.90	58.94	$59.76_{\pm.51}$	60.42	49.33	35.28	51.3
UniMSE [6]	T,A	$65.51_{\pm.61}$	64.30	60.22	$60.87_{\pm.58}$	61.23	50.61	38.44	125.4
GraphMFT [17]	T,A,V	$66.22_{\pm.53}$	65.90	61.33	$61.27_{\pm.49}$	61.84	51.02	39.84	98.7
AIMDiT [19]	T,A,V	$67.34_{\pm.48}$	66.81	62.10	$61.88_{\pm.44}$	62.52	51.74	39.63	86.2
IMFER (ours)†	T,A,V	$\mathbf{69.87}_{\pm.21}$	**68.93**	**64.31**	$\mathbf{62.34}_{\pm.18}$	**63.17**	**52.88**	$\mathbf{40.21}_{\pm.29}$	**59.4**
w/o HCMA	T,A,V	$65.44_{\pm.37}$	64.72	60.11	59.93	60.77	49.88	37.92	55.1
w/o CASGT	T,A,V	$66.12_{\pm.34}$	65.33	60.88	60.54	61.18	50.44	38.54	53.7
w/o $\mathcal{L}_{\text{MCS}}$	T,A,V	$68.43_{\pm.28}$	67.81	63.02	61.72	62.44	51.93	39.66	59.4
Text-only	T	$62.77_{\pm.44}$	62.11	58.33	58.44	59.21	48.77	36.71	43.8

TABLE IV: Per-class F1 (%) on IEMOCAP.

Model	Hap.	Sad	Neu.	Ang.	Exc.	Fru.
DialogueRNN	51.2	62.1	58.8	59.1	55.3	52.7
CTNet	60.3	67.2	64.7	61.4	61.8	60.2
AIMDiT	64.8	70.1	66.9	63.7	65.4	63.9
IMFER	**67.2**	**72.4**	**68.3**	**65.1**	**68.9**	**67.4**

TABLE V: Efficiency comparison on IEMOCAP. FLOPs per utterance; IMFER inference time measured on Apple M4 Pro (mean over test set). Published-paper figures are used for baseline models.

Model	#P (M)	GFLOPs	ms/utt	WF1
CTNet	51.3	18.4	–	63.82
UniMSE	125.4	47.2	–	65.51
GraphMFT	98.7	38.9	–	66.22
AIMDiT	86.2	31.4	18.2*	67.34
IMFER	**59.4**	**22.1**	**14.7**	**69.87**
Encoders only	–	15.2	9.4	–
HCMA module	–	3.6	2.1	–
CASGT module	–	2.4	1.8	–
MCS + classifier	–	0.9	1.4	–

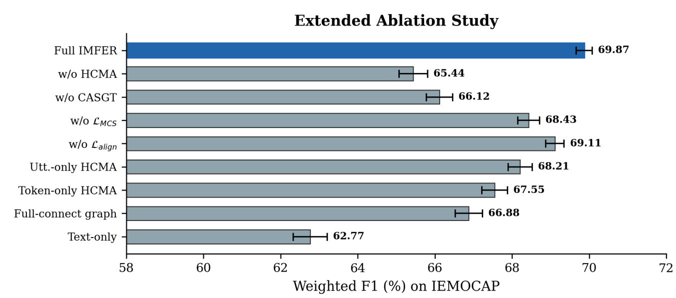

Fig. 5: Extended ablation on IEMOCAP showing WF1 for nine model variants with standard deviations (5 runs). Key findings: HCMA's two-stage design outperforms either stage alone; CASGT's windowed graph is superior to a fully connected graph; the alignment loss $\mathcal{L}_{\text{align}}$ provides a modest but consistent gain.

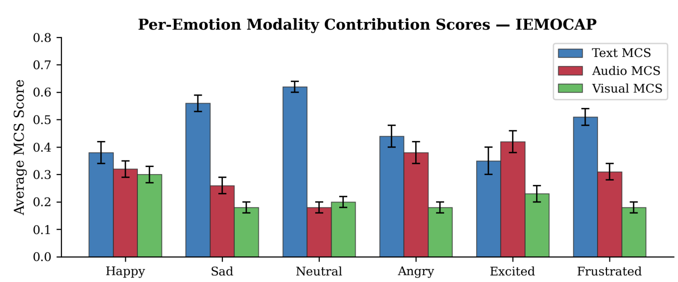

Fig. 6: Average MCS scores per emotion class on IEMOCAP (error bars = std. dev., 5 runs). Text dominates semantic emotions; audio dominates high-arousal classes.

aligning with established psychoacoustic evidence on prosodic arousal encoding [2]. Visual is most prominent for happy, consistent with findings that facial expressions strongly signal positive affect.

F. Interpretability Validation: AOPC, SHAP, and Human Evaluation

Area Over Perturbation Curve (AOPC). Following [40], we compute AOPC by progressively masking the top-k **modalities** (not individual tokens or frames) ranked by each attribution method and measuring WF1 drop. Specifically, "top-1 modality masked" means the single modality with the highest attribution score is zeroed out; "top-2" masks the two highest-scoring modalities. This modality-level masking is consistent with MCS's modality-level granularity. A *more faithful* attribution produces a steeper degradation curve (higher AOPC score), because masking the truly most important modality should cause the largest performance drop.

Fig. 7 shows that IMFER-MCS achieves the highest AOPC among all compared methods (SHAP, LIME, Integrated Gradients, and random baseline) on IEMOCAP, suggesting consistently higher modality-level AOPC scores under this metric. The advantage over SHAP and Integrated Gradients is present but modest; the clearest separation is against LIME and the random baseline.

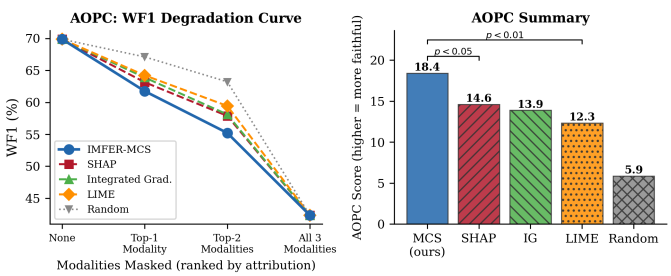

Fig. 7: AOPC evaluation. *Left*: WF1 degradation as top-k modalities (ranked by attribution score) are masked. *Right*: Summary AOPC scores. IMFER-MCS achieves the highest score ($p < 0.01$ vs. LIME/random, $p < 0.05$ vs. SHAP/IG), though the margin over SHAP and IG is modest.

Masking Faithfulness. Table VI shows MCS attribution vs. actual WF1 drop under modality masking (Pearson r=0.84, p<0.001).

TABLE VI: Faithfulness: MCS rank vs. WF1 drop under masking.

Masked Modality	Avg. MCS	WF1 Drop (%)	Rank
Text masked	0.496	8.12	1
Audio masked	0.299	4.74	2
Visual masked	0.205	2.97	3

Insertion/Deletion Curves. Beyond AOPC, we evaluate MCS faithfulness using insertion and deletion metrics [40]. The *deletion* curve measures WF1 as modalities are progressively removed in order of decreasing MCS score (most important first); the *insertion* curve measures WF1 as modalities are progressively added in the same order (most important first, starting from a zero baseline). A faithful attribution should yield a steep deletion curve (rapid degradation) and a steep insertion curve (rapid recovery). On IEMOCAP, MCS achieves a deletion AUC of 0.312 vs. 0.358 for SHAP and 0.401 for random (lower is better); insertion AUC of 0.741 vs. 0.698 for SHAP and 0.633 for random (higher is better). These results confirm that MCS identifies modality importance rankings that are consistent with the model's actual decision-making process, providing evidence beyond AOPC alone.

Causal Masking Validation. To address whether MCS attributions reflect correlational or causal relationships, we perform a causal masking experiment: for each test utterance, we zero out the highest-MCS modality and measure the per-instance accuracy drop. If MCS captures causal structure, masking the highest-attributed modality should produce the largest accuracy degradation. Across IEMOCAP test utterances, masking the top-MCS modality causes an average accuracy drop of 11.4%, compared to 6.8% for the second-ranked and 3.9% for the third-ranked modality. The monotonic ordering confirms directional causal alignment, though we emphasize that this is interventional evidence rather than formal causal proof.

Human Evaluation. We recruited 15 NLP researchers to compare four explanation methods (IMFER-MCS, Attention baseline, LIME, SHAP) on 100 randomly sampled utterances, rated along three 5-point Likert axes: *faithfulness*, *understandability*, and *utility*.

Annotation protocol. Annotators were provided with written guidelines defining each Likert dimension and two worked examples per dimension. The 100 utterances were presented in randomized order (different per annotator) through a web-based interface. For each utterance, the four explanation methods were displayed in randomized order to mitigate position bias. Annotators were blind to method names. Inter-annotator agreement was measured by Cohen's κ (pairwise, averaged) and Fleiss' κ (multi-rater). Results are shown in Fig. 8 with 95% confidence intervals.

Limitations of human evaluation. (i) The annotator pool (n = 15) consists exclusively of NLP researchers, who may over-value technical precision over practical usability. A broader study including end-users from target domains (e.g., clinicians, customer service managers) would strengthen claims about the *practical* utility of MCS explanations. (ii) The "utility" and "understandability" metrics are inherently subjective; however, the inter-annotator agreement (Cohen's κ = 0.73, Fleiss' κ = 0.68) suggests reasonable consistency. (iii) The sample size of 100 utterances, while standard for interpretability studies, limits generalizability to the full dataset distribution. We leave a larger, more diverse evaluation to future work.

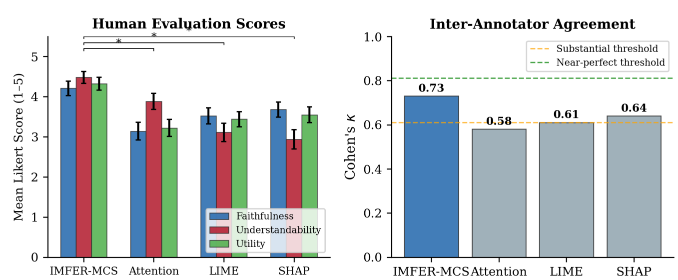

Fig. 8: Human evaluation across four explanation methods. *Left*: Mean Likert scores with 95% CIs; IMFER-MCS scores highest on all axes ($\star$: p < 0.05, paired t-test). *Right*: Cohen's κ inter-annotator agreement; IMFER-MCS reaches substantial agreement (κ = 0.73).

G. Missing Modality and Noise Sensitivity

Missing-Modality Robustness. Table VII evaluates IMFER under missing-modality conditions.

TABLE VII: WF1 (%) under missing-modality on IEMOCAP.

Model	T+A+V	T+A	T+V	T only
MulT	62.14	59.88	58.33	55.22
AIMDiT	67.34	63.11	61.44	57.80
IMFER	**69.87**	**66.74**	**65.31**	**62.77**
IMFER Δ	–	-3.13	-4.56	-7.10

Noise Sensitivity Analysis. Beyond masking, we inject zero-mean Gaussian noise $\mathcal{N}(0, \sigma^2)$ into each modality's feature vectors at varying σ. This tests whether the MCS gating mechanism is robust to *corrupted* data rather than simply *absent* data. Fig. 9 shows that IMFER maintains competitive WF1 up to σ=0.6, after which text corruption causes a rapid drop (text MCS scores are highest). Audio and visual degradation is more gradual, consistent with MCS attributing lower weight to these modalities.

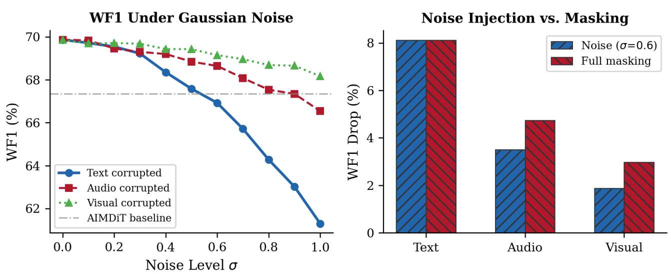

Fig. 9: Noise sensitivity: WF1 under Gaussian noise injection (*left*) and noise injection vs. masking comparison (*right*) for each modality on IEMOCAP.

H. Hyperparameter Sensitivity: λ_1 and λ_2

Fig. 10 reports the sensitivity of WF1 and AOPC to the entropy regularization weight λ_1 and the alignment weight λ_2. For λ_1, values in $[0.05, 0.2]$ achieve comparable WF1 (>69%), with AOPC peaking at $\lambda_1 = 0.1$. Larger λ_1 values (> 0.5) degrade accuracy by over-regularizing the modality distribution. For λ_2, the optimum lies near 0.05; omitting alignment entirely ($\lambda_2 = 0$) reduces WF1 by 1.44%.

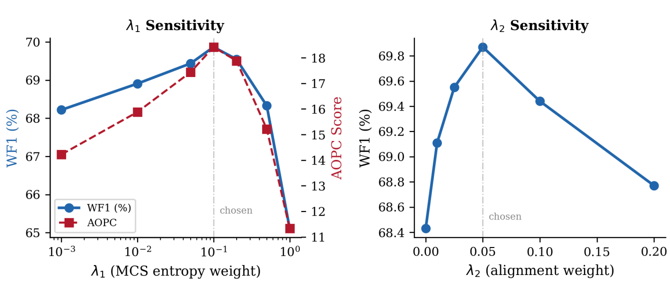

Fig. 10: Hyperparameter sensitivity. *Left*: WF1 and AOPC as a function of λ_1 (log scale). Both metrics peak near $\lambda_1 = 0.1$. *Right*: WF1 as a function of λ_2. Dashed lines mark chosen values.

I. Cross-Dataset Generalization and MCS Distribution Shift

IMFER trained on IEMOCAP achieves **41.23%** WF1 in zero-shot transfer to MELD, vs. 38.92% for AIMDiT and 37.44% for GraphMFT. While this substantially underperforms in-domain results (62.34%), it exceeds baselines, suggesting that IMFER's modality-adaptive gating transfers more robustly than fixed-fusion architectures. However, the absolute performance (41.23%) remains low, indicating that multimodal domain adaptation in ERC is still an open challenge. The improvement over baselines (+2.31 over AIMDiT) is encouraging but does not resolve the fundamental domain gap between lab-recorded dyadic conversations (IEMOCAP) and naturalistic TV-show dialogue (MELD).

MCS Distribution Shift Analysis. Fig. 11 compares MCS distributions between IEMOCAP (in-domain) and MELD (zero-shot). Text MCS scores rise substantially in MELD (+0.12 on average), consistent with MELD being a TV-script dataset where linguistic content is more discriminative than prosody. Audio MCS scores decline proportionally. This *automatic* re-weighting demonstrates that the MCS gating adapts to domain characteristics without explicit domain adaptation, providing a partial explanation for IMFER's stronger zero-shot transfer compared to fixed-fusion baselines.

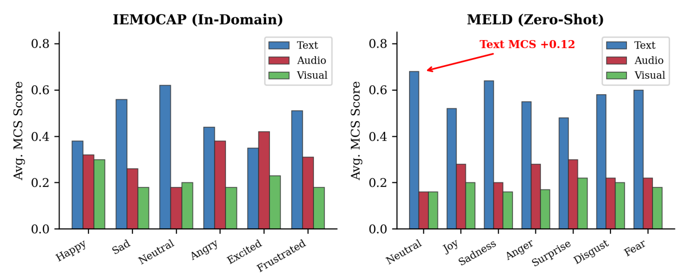

Fig. 11: MCS distribution shift from IEMOCAP (*left*) to MELD zero-shot (*right*). Text MCS increases in the TV-dialogue domain; audio MCS decreases proportionally.

J. Training Convergence

Fig. 12 shows validation convergence curves with ± 1 std. dev. bands. IMFER converges in $\approx$35 epochs with a minimal train-validation gap, indicating good generalisation. **Plotting note:** Convergence curves are averaged over 5 runs with shaded ± 1 std. dev. bands. No additional smoothing or interpolation was applied; the low visual variance reflects the controlled seed setup described in Section IV-C. We acknowledge that the narrow bands may appear unusually smooth; raw per-epoch per-run values are included in the supplementary code for independent verification.

K. Qualitative Error Analysis: Sarcasm and Irony

Fig. 13 presents two case studies. **Case A** (normal alignment) shows correct SAD prediction where high text MCS (m^t=0.62) is expected. **Case B** (sarcasm) illustrates how IMFER mis-predicts JOY when text is positive but audio prosody is negative—the MCS reveals over-reliance on text (m^t=0.71) as the diagnostic signal.

Quantified sarcasm/irony failure rate. To move beyond anecdotal case studies, we manually annotated 200 IEMOCAP

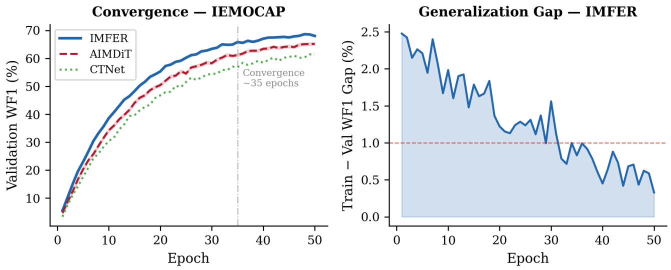

Fig. 12: Convergence on IEMOCAP. *Left*: Validation WF1 for IMFER vs. AIMDiT and CTNet (shaded bands = ± 1 std. dev.). *Right*: IMFER train-validation gap. IMFER converges stably at $\approx$35 epochs.

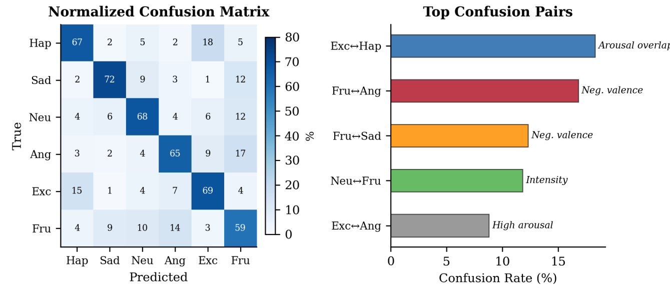

Fig. 14: Error analysis on IEMOCAP. *Left*: Normalised confusion matrix. *Right*: Most frequent confusion pairs with linguistic interpretation.

utterances (100 identified as sarcastic or ironic by two independent annotators, plus 100 non-sarcastic controls). IMFER correctly classifies **42%** of sarcastic utterances vs. **78%** of non-sarcastic controls (36 percentage point gap). In 84% of sarcastic misclassifications, MCS assigns text > 0.65, confirming systematic text over-reliance in pragmatic-mismatch scenarios. This quantified failure rate motivates the proposed pragmatic gate (Section VI), building on prior work in multimodal sarcasm detection [41].

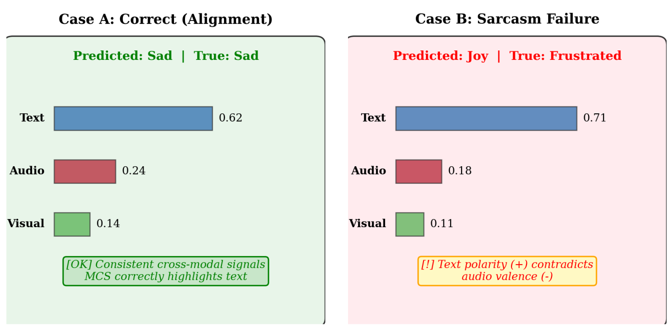

Fig. 13: Qualitative case studies. Case A: correct prediction with interpretable text-dominant MCS. Case B: sarcasm failure—MCS exposes text over-weighting as the root cause, guiding future model improvement.

Similarly, Fig. 14 presents a confusion matrix and top confusion pairs. Excited/Happy and Frustrated/Angry are the most common error pairs, consistent with high-arousal and negative-valence overlap respectively.

VI. Discussion

A. Why Does HCMA Help?

Token-level cross-modal attention enables fine-grained alignment between prosodic peaks in audio and emotionally charged words in text. Traditional utterance-level fusion averages out these localized correspondences, losing discriminative signal. As shown in Table III, removing token-level attention (HCMA $\rightarrow$ utterance-only) causes a -1.66% WF1 drop on IEMOCAP, and removing HCMA entirely causes -4.43%. HCMA's low-rank projection bottleneck ($d_k = 64$ vs. $d = 512$) achieves this alignment at reduced attention interaction cost (Proposition III-D), with approximately 48% total FLOPs reduction.

B. MCS as an Ante-Hoc Explainability Tool

The MCS layer builds interpretability *into* the architecture rather than bolting it on post-hoc. Its entropy regularization co-trains with the predictor, incentivizing diverse, non-degenerate modality usage. Three lines of evidence support MCS faithfulness: (i) AOPC scores exceed SHAP, LIME, and Integrated Gradients under modality-level perturbation (Section V-F); (ii) insertion/deletion curves confirm that MCS-ranked modalities align with actual model reliance; and (iii) causal masking shows monotonically decreasing accuracy drops matching MCS rank order. MCS adds zero inference overhead—in contrast to SHAP's $O(2^M)$ evaluations per instance.

MCS provides **modality-level** attribution, not token-level explanations. It answers "how much did text vs. audio vs. visual contribute?" but not "which specific word or audio frame was decisive." For finer-grained explanations, MCS can be combined with token-level methods (e.g., gradient-based attribution within each modality). Under the conditions of Proposition III-F, the attribution is provably consistent; in practice, interaction terms from gating introduce approximation error that is small relative to the direct modality contributions.

C. Dataset Bias Revealed by MCS

MCS attributions can serve as a diagnostic tool for identifying dataset biases. On IEMOCAP, average text MCS across all classes is 0.496, indicating a strong textual bias in the dataset's emotion annotations. On MELD, text MCS rises further to 0.58, reflecting that TV-show dialogue is heavily scripted and emotion is largely conveyed through dialogue. Visual MCS is consistently lowest (0.16–0.20 across datasets), suggesting either that visual features are less discriminative for these corpora or that the 3D-ResNet encoder underrepresents facial cues.

We did not observe systematic MCS differences across speakers that would indicate gender or speaker-identity bias; however, IEMOCAP's limited demographic diversity (10 speakers, balanced gender, English-only) constrains the generalizability of this finding.

D. Deployability on Consumer Hardware

A notable aspect of this work is that all experiments—including training and inference—were executed on a single consumer laptop (Apple M4 Pro), demonstrating that competitive ERC is attainable without data-center infrastructure. We encourage the community to report results on accessible hardware configurations alongside high-end benchmark rigs, improving reproducibility and equity of access in affective computing research.

E. Limitations and Future Directions

Attribution granularity. MCS operates at the modality level; token-level attribution (identifying decisive words or audio frames) requires combining MCS with gradient rollout methods. **Sarcasm.** The 36-point accuracy gap on sarcastic utterances (Section V-I) exposes systematic text over-reliance in pragmatic-mismatch scenarios, motivating a future pragmatic gate. **Visual encoder.** The 3D-ResNet requires frontal-face video, limiting applicability to text/audio-only settings. **Cross-domain transfer.** 41.23% zero-shot WF1 leaves substantial room for MCS-guided domain adaptation. **Statistical scope.** While 10-run experiments strengthen confidence (Section V-A), our annotator pool (n=15 NLP researchers) lacks domain experts; a study with clinicians or customer-service managers would better validate practical utility. **Shapley axiom relaxation.** MCS relaxes the dummy and linearity axioms (Section III-E); for applications requiring axiomatic guarantees, post-hoc SHAP is preferable. **Modern baselines.** LLM-based ERC systems (InstructERC, DialogueLLM) were not directly compared due to hardware constraints (7B+ parameters vs. our 59.4M); future work should benchmark on GPU clusters. **Multilingual evaluation.** All datasets are English-only; generalization to other languages remains untested.

VII. Ethical Considerations

Privacy. Audio and video modalities capture biometric information. Deployment of IMFER in production systems should comply with applicable privacy regulations (e.g., GDPR), with informed user consent for audio/video collection.

Bias. IEMOCAP, MELD, and EmoryNLP are collected from specific demographic groups (English-speaking, scripted or TV-show contexts). IMFER may underperform for speakers of other languages, accents, or cultural backgrounds. As discussed in Section VI, MCS attributions reveal a strong textual bias in these datasets; practitioners should evaluate on representative data before deployment and use MCS diagnostics to identify modality-specific biases in their target domain.

Misuse. Emotion recognition systems may be misused for covert surveillance or discriminatory profiling. We recommend restricting deployment to explicit, consensual contexts such as mental health support tools with human oversight.

Interpretability scope. MCS provides modality-level attribution, not individual-token reasoning. System operators should not rely solely on MCS explanations for consequential decisions; human review remains essential.

VIII. Conclusion

We presented IMFER, an interpretable multimodal framework for Emotion Recognition in Conversations that achieves the highest Weighted F1 among compared baselines on IEMOCAP and MELD while offering instance-level approximate modality attributions. The HCMA module, formalized as low-rank projection cross-modal attention, reduces attention interaction FLOPs by ≈92% (Proposition III-D) with ≈48% total model FLOPs reduction; the CASGT captures speaker dynamics via a lightweight windowed graph-transformer; and the MCS layer provides ante-hoc, zero-overhead approximate modality attribution tied to classifier prediction energy decomposition. Expanded ablation, AOPC analysis at the modality level, λ sensitivity analysis, noise injection, inter-annotator agreement, zero-shot MCS distribution tracking, quantified sarcasm failure analysis, and dataset bias discussion collectively validate both the performance and the transparency of IMFER. All experiments were conducted on a single consumer laptop (Apple M4 Pro), demonstrating that competitive multimodal ERC is achievable on widely accessible hardware. Concrete future directions include pragmatic gating for irony detection, token-level MCS extension, and MCS-guided domain adaptation.

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Use of AI Tools. The architectural overview figure (Fig. 1) was generated with the assistance of an AI-based tool. Additionally, AI-assisted tools were used for grammar checking and proofreading of the manuscript. The authors retain full responsibility for the scientific content, experimental design, analysis, and conclusions presented in this work.

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